HEAPLINE · USER REQUIREMENTS

HeapLine user requirements

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-11-03	Amish Chadha	Initial issue (kit 1.9.1 upgrade); needs taken from the users and stakeholders in HPL-PRB-001 Table 1 and from the requirements decided on 2026-10-03 and 2026-10-04

HeapLine user requirements

What the people who would use or stock HeapLine need, written from their side. The product requirements that meet these needs are in the PRS (`03-requirements.md`) and the links are in the traceability matrix (`03b-traceability.md`). Method: STANDARDS section 22.

Table 1. User requirements.

ID	Category	Title	Need	Priority	Source
UR-001	Deployment and assembly	First search line fast	In the first minutes after a slide, the people on site can fetch the kit and start searching before formal responders arrive	High	Waste pickers and cooperatives (PRB Table 1)
UR-002	Performance	Find a buried person	Searchers can locate a buried colleague at the depth people are typically buried, from the surface	High	Waste pickers and cooperatives; responders (PRB Table 1)
UR-003	Performance	Work on loose waste	Searchers can move over loose, voided debris without sinking or falling through	High	Waste pickers and cooperatives (PRB Table 1)
UR-004	Performance	Clear heavy cover	Sheeting, timber or heavy debris lying over a mark can be moved when hands alone cannot	High	Waste pickers and cooperatives (PRB Table 1)
UR-005	Performance	Move a casualty	A casualty can be moved over debris to a pickup point by the people on site	High	Waste pickers; neighbours (PRB Table 1)
UR-006	Usability	A clear point to stop	Every searcher knows the stop signals and the safe zone before stepping onto the debris	High	Neighbours; responders (PRB Table 1)
UR-007	Environment and durability	Ready after months in a shed	The kit works on the day after months stored in heat, damp and dust	High	Site operators and municipal staff (PRB Table 1)
UR-008	Logistics	Carried by few people	The kit reaches the slide carried by four people or fewer, with nothing too heavy for one person	High	Waste pickers and cooperatives (PRB constraints)
UR-009	Cost	Affordable to stock	A site or a cooperative can afford to stock and inspect the kit	High	Site operators and municipal staff (PRB Table 1)
UR-010	Safety	Does not add danger	The equipment cannot make a bad situation worse: no rope overload, no drum run-back, nothing that pulls on a person	High	All users (PRB safety note); decisions 2, 3 and 6 of HPL-DDR-001
UR-011	Maintenance	Repairable locally	The kit can be inspected monthly and repaired with common steel, timber, rope and tools	Medium	Site operators and municipal staff (PRB constraints)
UR-012	Usability	Hand over an organised search	Responders arrive to a search with a lookout, marked burial points and a clear handover	Medium	Fire, police and Red Cross responders (PRB Table 1)
UR-013	Regulatory	Not mistaken for certified equipment	The kit is published and described as an open engineering reference, not certified rescue equipment	Medium	Site operators and municipal staff; responders (PRB safety note)

ID	Category	Title	Need	Priority	Source
UR-014	Performance	Work in the dark and rain	Searchers can see and be seen at night and in rain, without power at the scene	Medium	Waste pickers and cooperatives (PRB operating environment)
UR-015	Logistics	A shared capstan arrives in time	Where neighbouring sites share one capstan set, it arrives while the search is still under way	Medium	Site operators; decisions 23C and 3A (HPL-DDR-003, HPL-DDR-004)

Categories: deployment and assembly, performance, usability, safety, environment and durability, logistics, maintenance, cost, regulatory. Priorities: High, Medium, Low.